

Family Identity & Beneficiary Management Platform

Product Requirements Document

Version 1.0

Project Type:	Government Digital Platform
Document Type:	Product Requirements Document
Version:	1.0
Status:	Proposed
Target Users:	Citizens and Government Officers

Contents

1	Executive Summary	3
2	Problem Statement	3
3	Product Vision	3
4	Product Objectives	4
5	Target Users	4
5.1	Citizens	4
5.2	Government Officers	5
5.3	System Administrator	5
6	Core Product Concept	5
7	Family Registration	6
8	Family ID Management	6
9	Family Tree Management	7
10	Relationship Verification	8
11	Duplicate Detection	8
11.1	Exact Matching	8
11.2	Fuzzy Matching	9
12	Family Lifecycle Management	9
13	Beneficiary Management	10
14	Family 360-Degree View	10
15	Officer Verification Workflow	11
16	Audit Trail	12
17	Notification System	13
18	Functional Requirements	13

19 Non-Functional Requirements	14
19.1 Security	14
19.2 Performance	15
19.3 Scalability	15
19.4 Availability	15
19.5 Maintainability	15
20 Role-Based Access Control	15
21 High-Level Data Model	16
21.1 Family	16
21.2 Family Member	17
21.3 Relationship	17
22 Security and Privacy	17
23 High-Level System Architecture	18
24 Out of Scope for MVP	19
25 Minimum Viable Product	19
26 Future Enhancements	20
27 Success Metrics	20
28 Risks and Mitigation	21
29 Key Hackathon Differentiators	21
30 Example User Journey	22
31 Conclusion	23

1 Executive Summary

The Family Identity and Beneficiary Management Platform is a centralized digital platform designed to create and maintain a trusted family-level identity for citizens.

The platform enables citizens to register their family, verify individual members, establish relationships through supporting documents, and receive a unique Family ID.

The Family ID acts as a common reference for connecting eligible family members with government welfare schemes and beneficiary programs.

The system also provides government officials with tools for verification, duplicate detection, family lifecycle management, beneficiary tracking, analytics, and audit management.

2 Problem Statement

Government welfare programs often require information about family members, their relationships, residence, and eligibility.

In a fragmented system, the same family may have to provide similar information repeatedly for different government schemes.

This can result in:

- Duplicate beneficiary records.
- Repeated document verification.
- Inconsistent family information.
- Difficulty verifying relationships.
- Delays in beneficiary approval.
- Difficulty tracking family-level benefits.
- Lack of a unified family view for government officers.

The proposed platform addresses these problems by maintaining a verified family profile that can be referenced across multiple government services.

3 Product Vision

The vision of the platform is:

“Create a trusted digital family identity that enables secure, verified, and efficient delivery of government services.”

The system should provide a single family-level view while maintaining individual identity and relationship information.

4 Product Objectives

The primary objectives are:

1. Create a unique Family ID for every verified family.
2. Digitally maintain family relationships.
3. Verify family members using approved identity documents.
4. Reduce duplicate family and beneficiary records.
5. Provide government officers with a Family 360-degree view.
6. Connect families with applicable government schemes.
7. Maintain a complete audit history of important changes.
8. Support family lifecycle events such as birth, marriage, migration, and death.
9. Reduce repetitive document submission and verification.
10. Improve transparency and efficiency in beneficiary management.

5 Target Users

5.1 Citizens

Citizens can:

- Register a family.
- Add family members.
- Submit identity documents.
- Submit relationship documents.
- View family information.
- Request corrections.
- View scheme eligibility and applications.

5.2 Government Officers

Government officers can:

- Verify family registrations.
- Verify relationships.
- Review submitted documents.
- Resolve duplicate records.
- Approve or reject requests.
- View family information.
- Manage beneficiary records.
- Generate reports.

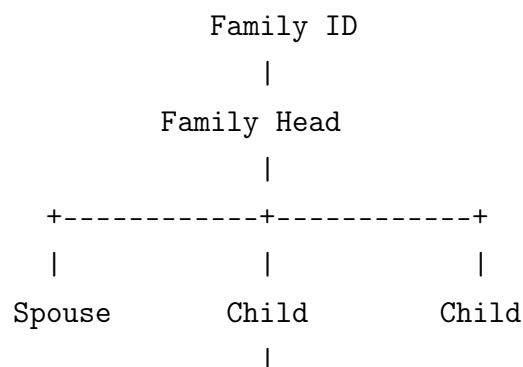
5.3 System Administrator

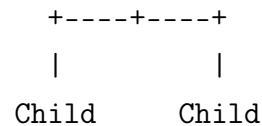
Administrators can:

- Manage users and roles.
- Configure schemes.
- Manage system settings.
- Monitor system activity.
- Access audit logs.

6 Core Product Concept

The core concept is a hierarchical family identity structure.





Each person retains an individual identity while being associated with a Family ID.

The Family ID should remain stable even if the family changes its address.

Geographical information should therefore be stored as metadata rather than permanently encoded into the identity itself.

7 Family Registration

The family registration workflow consists of:

1. Citizen initiates family registration.
2. Family Head provides identity information.
3. Identity is verified using an approved identity mechanism.
4. Existing family records are searched.
5. If no matching family exists, a new family record is created.
6. A unique Family ID is generated.
7. Family members are added.
8. Relationships are established.
9. Supporting documents are submitted where required.
10. Government officer verifies the registration.
11. Family becomes an officially verified family record.

8 Family ID Management

Every verified family receives a unique Family ID.

Example:

GJ-FAM-7A82K9X4

The Family ID should:

- Be globally unique.

- Remain stable throughout the family lifecycle.
- Not expose sensitive personal information.
- Not directly contain the citizen's identity number.
- Be generated using a secure identifier-generation mechanism.

Location information should be stored separately:

Family ID

```
|
+-- State
+-- District
+-- Taluka
+-- Village/Ward
+-- Current Address
```

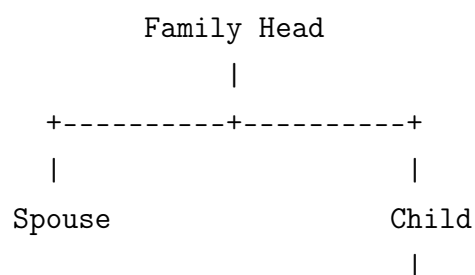
9 Family Tree Management

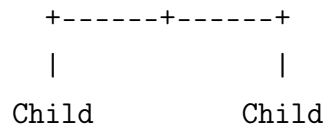
The platform shall provide a visual family tree.

Each family member should contain:

- Member ID.
- Name.
- Date of birth.
- Gender.
- Relationship.
- Verification status.
- Current status.

Example:





The tree should support dynamic updates when family relationships change.

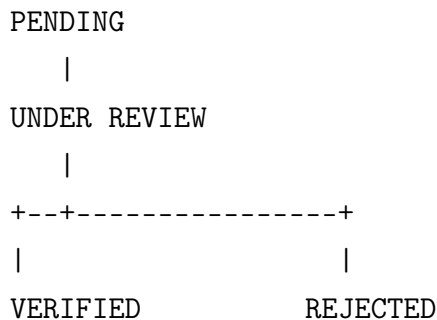
10 Relationship Verification

The system shall support relationship verification.

Possible supporting documents include:

- Ration Card.
- Birth Certificate.
- Marriage Certificate.
- Other approved government documents.
- Affidavit where appropriate.

Each relationship should have a verification status.



11 Duplicate Detection

The platform shall identify potentially duplicate citizen and family records.

Duplicate detection may use:

11.1 Exact Matching

- Government identity reference.
- Mobile number.
- Date of birth.
- Existing Family ID.

11.2 Fuzzy Matching

- Name similarity.
- Address similarity.
- Parent or spouse name.
- Other permitted demographic information.

The system should not automatically merge records solely based on similarity. Potential duplicates should be sent to an authorized officer for review.

12 Family Lifecycle Management

Families change over time.

The platform shall support:

- Birth / addition of a new member.
- Marriage.
- Divorce or separation.
- Death.
- Migration.
- Address change.
- Relationship correction.
- Family restructuring.

Historical records should be retained.

Instead of deleting a member:

Member Status:

ACTIVE

DECEASED

MIGRATED

SEPARATED

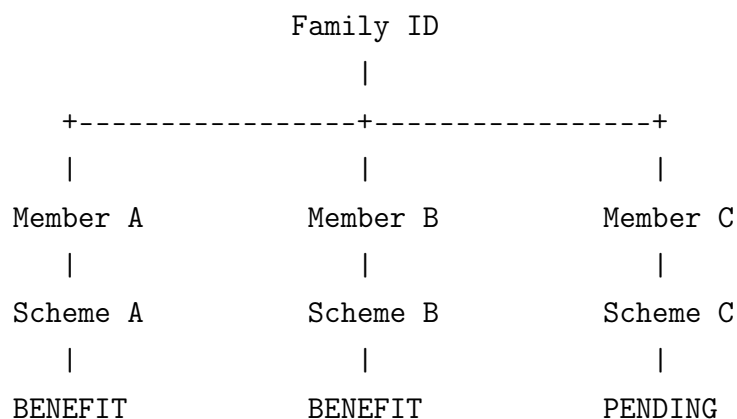
INACTIVE

This preserves historical integrity.

13 Beneficiary Management

The Family ID can act as a common reference for beneficiary management.

Example:



Government officers can view:

- Active benefits.
- Previous benefits.
- Pending applications.
- Scheme eligibility.
- Beneficiary status.

14 Family 360-Degree View

Authorized government officers shall have access to a consolidated Family 360-degree dashboard.

The dashboard should contain:

- Family ID.
- Family Head.
- Location.
- Family members.
- Family tree.
- Verification status.

- Submitted documents.
- Scheme information.
- Beneficiary information.
- Pending requests.
- Audit history.

Example:

+-----+	
	FAMILY 360
+-----+	
	Family ID: GJ-FAM-7A82K9X4
	Family Head: Example Person
	Members: 5
	Verification: 5/5
+-----+	
	FAMILY TREE
	Family Head
	+-----+
	Spouse Child
+-----+	
	BENEFITS
	Scheme A
	Scheme B
	Pending: Scheme C
+-----+	

15 Officer Verification Workflow

The officer dashboard shall provide a verification queue.

Example:

ID	Request	Status	Action
1023	New Family Registration	Pending	Review

1024	Relationship Verification	Pending	Review
1025	Possible Duplicate	High Priority	Investigate
1026	Address Change	Pending	Review

Officers can:

- Approve.
- Reject.
- Request additional documents.
- Flag a duplicate.
- Escalate a case.

16 Audit Trail

All sensitive changes must be recorded.

The audit record should contain:

- User/Officer ID.
- Timestamp.
- Operation.
- Record affected.
- Previous value.
- New value.
- Reason.

Example:

Officer: OFF-1023

Date: 20-09-2026

Changed:

Relationship

Old Value:

Brother

New Value:

Father

Reason:

Approved supporting document

17 Notification System

The system may notify citizens about:

- Registration status.
- Document verification.
- Relationship verification.
- Duplicate review.
- Scheme application status.
- Required additional documents.
- Important family record changes.

Notifications may be delivered through approved channels such as:

- SMS.
- Email.
- In-application notifications.

18 Functional Requirements

ID	Requirement
FR-01	The system shall allow citizens to register a family.
FR-02	The system shall allow registration of a Family Head.
FR-03	The system shall generate a unique Family ID.
FR-04	The system shall allow members to be added to a family.
FR-05	The system shall maintain family relationships.

FR-06	The system shall support document submission.
FR-07	The system shall support relationship verification.
FR-08	The system shall provide a visual family tree.
FR-09	The system shall detect potentially duplicate records.
FR-10	The system shall provide an officer verification queue.
FR-11	The system shall support approval and rejection workflows.
FR-12	The system shall allow officers to request additional documents.
FR-13	The system shall maintain family lifecycle information.
FR-14	The system shall support beneficiary management.
FR-15	The system shall map families to government schemes.
FR-16	The system shall provide a Family 360-degree view.
FR-17	The system shall maintain an audit trail.
FR-18	The system shall support role-based access control.
FR-19	The system shall generate administrative reports.
FR-20	The system shall provide notifications for important events.
FR-21	The system shall support address changes.
FR-22	The system shall preserve historical records.
FR-23	The system shall allow authorized officers to resolve duplicates.
FR-24	The system shall provide dashboard analytics.
FR-25	The system shall protect sensitive citizen information.

19 Non-Functional Requirements

19.1 Security

The system shall:

- Use encrypted communication.
- Implement role-based access control.
- Protect sensitive information.
- Maintain audit logs.
- Use secure authentication.
- Follow data minimization principles.

19.2 Performance

The system should:

- Support concurrent users.
- Provide dashboard responses within acceptable time.
- Efficiently search family records.
- Process duplicate detection asynchronously when required.

19.3 Scalability

The architecture should support expansion from:

Village
↓
Taluka
↓
District
↓
State
↓
Multiple States

19.4 Availability

The system should be designed for high availability because government services may require continuous access.

19.5 Maintainability

The platform should use modular services so that identity verification, family management, beneficiary management, notification, and analytics can evolve independently.

20 Role-Based Access Control

Role	Permissions
Citizen	Manage own family information, submit documents, view requests.

Verification Officer	Review documents, verify relationships, approve/reject requests.
District Officer	Review district-level families and reports.
Administrator	Manage users, schemes, configuration, and audit information.
System Auditor	Access audit records and compliance information.

21 High-Level Data Model

The primary entities are:

FAMILY

```

|
+---- FAMILY_MEMBER
|
+---- FAMILY_RELATIONSHIP
|
+---- DOCUMENT
|
+---- BENEFICIARY
|
+---- SCHEME_APPLICATION
|
+---- AUDIT_LOG
|
+---- ADDRESS_HISTORY

```

Important entities include:

21.1 Family

- Family ID.
- Family Head.
- Location.
- Registration Date.
- Verification Status.

21.2 Family Member

- Member ID.
- Family ID.
- Name.
- Date of Birth.
- Gender.
- Status.

21.3 Relationship

- Relationship ID.
- Source Member.
- Target Member.
- Relationship Type.
- Verification Status.

22 Security and Privacy

The system should follow a privacy-by-design approach.

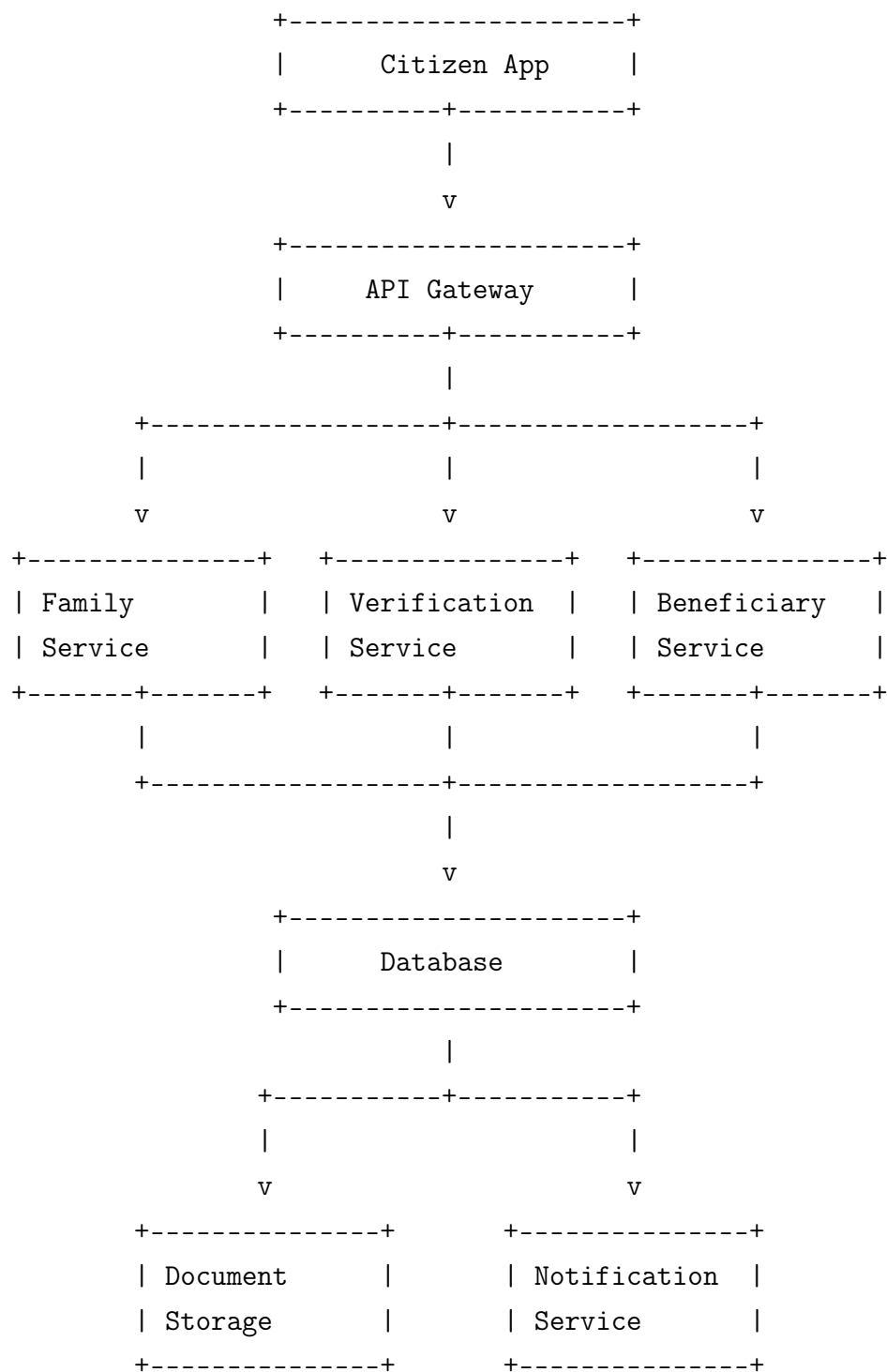
Security measures include:

- Encryption in transit.
- Encryption at rest.
- Role-based authorization.
- Strong authentication.
- Audit logging.
- Data minimization.
- Masking of sensitive identifiers.
- Secure document storage.
- Access logging.

Raw identity numbers should not be unnecessarily exposed or displayed through the application.

For a hackathon prototype, only synthetic/dummy identity data should be used.

23 High-Level System Architecture



24 Out of Scope for MVP

The following features are outside the initial MVP:

- Full integration with every government department.
- Real Aadhaar database integration.
- Real financial transactions.
- Automatic legal determination of relationships.
- Nationwide deployment.
- Replacement of existing government identity systems.

These can be considered for future phases.

25 Minimum Viable Product

The hackathon MVP should contain:

1. Citizen registration.
2. Family Head registration.
3. Family ID generation.
4. Family member management.
5. Family tree visualization.
6. Document upload.
7. Relationship verification.
8. Duplicate detection prototype.
9. Officer dashboard.
10. Scheme mapping.
11. Beneficiary status.
12. Audit trail.

26 Future Enhancements

Potential future features include:

- Mobile application.
- Multilingual interface.
- Offline government-office support.
- AI-assisted duplicate detection.
- Automated document classification.
- Advanced fraud detection.
- Inter-department data integration.
- Analytics and predictive dashboards.
- API integration with authorized government systems.

27 Success Metrics

The platform can measure:

- Percentage of verified family records.
- Average family registration time.
- Average verification time.
- Number of duplicate records detected.
- Number of duplicate records resolved.
- Percentage of digitally verified relationships.
- Beneficiary application processing time.
- Number of families connected to schemes.
- Number of successful citizen requests.

28 Risks and Mitigation

Risk	Impact	Mitigation
Incorrect relationship	High	Multi-document verification and officer approval.
Duplicate family	High	Exact and fuzzy duplicate detection.
Unauthorized access	High	RBAC, authentication and audit logs.
Incorrect data	Medium	Verification workflow and correction requests.
Data breach	High	Encryption, access control and secure storage.
System downtime	High	High availability and backup mechanisms.

29 Key Hackathon Differentiators

The solution differentiates itself through:

1. **Family 360-Degree View**

A unified view of family members, relationships, documents, verification, and benefits.

2. **Relationship Graph**

Family information is represented as a graph instead of only traditional tabular records.

3. **Duplicate Detection**

Potential duplicate families and members are automatically flagged.

4. **Family Lifecycle**

Birth, marriage, migration, death, and other changes are maintained historically.

5. **Beneficiary Intelligence**

Family information can be used to identify relevant government scheme relationships.

6. **Auditability**

Every important modification has a traceable history.

30 Example User Journey

```
Citizen
|
v
Login
|
v
Register Family
|
v
Verify Identity
|
v
Add Family Members
|
v
Upload Relationship Documents
|
v
Submit for Verification
|
v
Government Officer Review
|
+---- Reject ----> Additional Documents
|
+---- Approve
      |
      v
      Family ID
      |
      v
      Family Verified
      |
      v
      Scheme Mapping
      |
      v
```

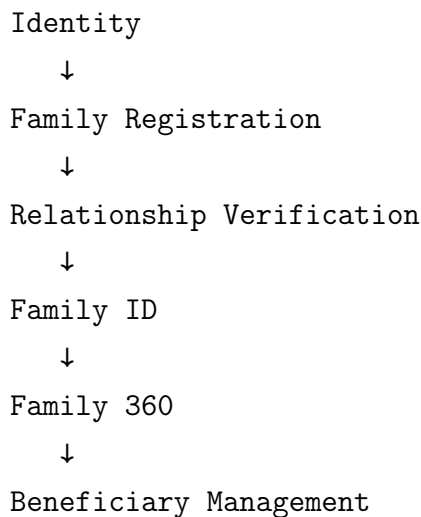
Beneficiary Services

31 Conclusion

The Family Identity and Beneficiary Management Platform provides a centralized approach to maintaining verified family information.

By combining family identity, relationship management, duplicate detection, beneficiary management, lifecycle tracking, and auditability, the platform can provide government departments with a reliable family-level information layer while reducing repetitive verification processes for citizens.

The proposed MVP focuses on demonstrating the core workflow:



The architecture is designed so that future integrations with authorized government systems can be added without redesigning the core family management model.